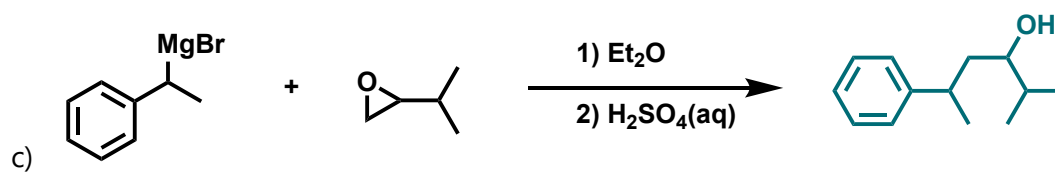
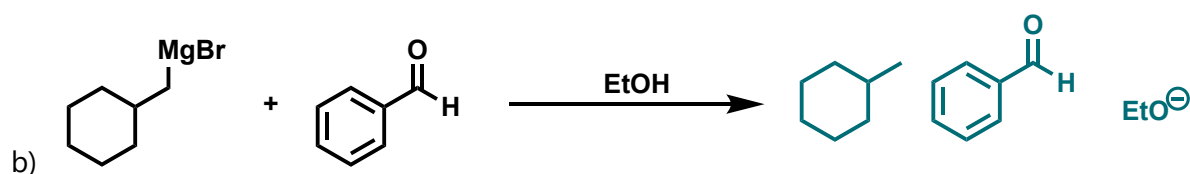
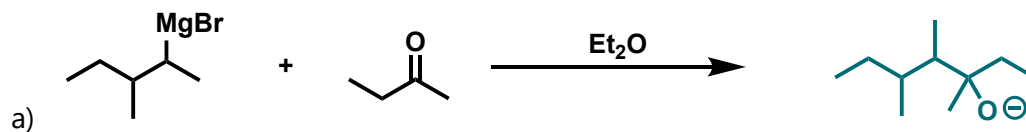


Chem 0345/0330 – Organic Lab

Activity 7: Grignard Reaction

Extra Practice Problems - KEY

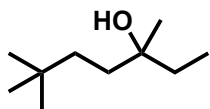
1. Predict the product for the following reactions:



After making the Grignard reagent, an intramolecular cyclization will occur:



2. Choose which sets of reactants could be used to synthesize the product shown below. More than one set of reactants is correct for each product. Assume the brominated reagent is reacted with magnesium, then combined with the shown electrophile followed by an acidic workup.



- a) CC(C)(C)CCBr + CC(=O)CC >> CC(C)(C)CC(O)(C)CC
- b) CC(C)C(C)CBr + CC(=O)CC >> CC(C)(C)CC(O)(C)CC
- c) CC(C)(C)CCBr + CC1(C)OC1 >> CC(C)(C)CCCCO
- d) CC(C)CCBr + CCOC1C >> CC(C)(C)CC(O)(C)CC
- e) CCCBr + CC(C)(C)CC=O >> CC(C)(C)CC(O)(C)CC
- f) CCCBr + CC(C)(C)CC(=O)C >> CC(C)(C)CC(O)(C)CC
- g) CCCBr + CC(C)(C)CC=O >> CC(C)(C)CC(O)(C)CC

3. Choose which sets of reactants could be used to synthesize the product shown below. More than one set of reactants is correct for each product. Assume the brominated reagent is reacted with magnesium, then combined with the shown electrophile followed by an acidic workup.

